CBSE Mathematics 2015

Class X – Outside Delhi, Code 30/1



Formula-first worked solutions with matrix methods

SCOPE AND SOURCE AUDIT

This build intentionally covers only the Class X Mathematics paper with code 30/1. The official CBSE paper has 31 questions: Section A (Q1–4), Section B (Q5–10), Section C (Q11–20), and Section D (Q21–31). Class XII is left out of this build until the Class X layout and verification pass are finalized.

The solutions state the general formulae first and refer back to them when solving individual questions. Coordinate-geometry work is expressed with vectors, dot products, determinants, and matrix systems wherever that makes the reasoning clearer. The level of mathematics is deliberately enriched, but each solution still gives a short path to the requested school-level result. Comments in the source files explain why a particular method is selected; they are mathematical teaching notes rather than an unfiltered private reasoning trace.

Primary source: official CBSE 2015 paper 30/1 and the corresponding marking scheme. The source audit records transcription mismatches in `STRUCTURE_AUDIT.md`.

CONTENTS

1 Class X Mathematics – Code 30/1 4
1.1 Formulae 4
1.2 Questions and worked solutions 7
1.2.1 Section A – 1 mark each 7
1.2.2 Section B – 2 marks each 8
1.2.3 Section C – 3 marks each 11
1.2.4 Section D – 4 marks each 16

1. Class X Mathematics – Code 30/1

A. Formulae

1. Quadratic equations and algebra: For $ax^2+bx+c=0$ with $a \neq 0$, the discriminant and roots are

$$\Delta = b^2 - 4ac, \qquad x = \frac{-b \pm \sqrt{\Delta}}{2a}. \tag{F1}$$

Equal real roots occur exactly when $\Delta = 0$. The condition $a \neq 0$ matters: if the leading coefficient vanishes, the equation is no longer quadratic.

The difference-of-squares identity is

$$u^2 - v^2 = (u - v)(u + v). \tag{F2}$$

It is useful to factor an expression before applying the quadratic formula when the factors are visible.

2. Right-triangle trigonometry and motion: For angle θ in a right triangle,

$$\tan\theta = \frac{\text{opposite}}{\text{adjacent}}, \quad \sin\theta = \frac{\text{opposite}}{\text{hypotenuse}}, \quad \cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}}. \tag{F3}$$

The standard values used in this paper are $\tan 30^\circ = 1/\sqrt{3}$, $\tan 60^\circ = \sqrt{3}$, $\sin 30^\circ = 1/2$, and $\cos 60^\circ = 1/2$. For a right triangle with legs a, b and hypotenuse c,

$$c^2 = a^2 + b^2, \qquad \|(x, y)^T\| = \sqrt{x^2 + y^2}. \tag{F4}$$

The second identity is the Euclidean length of a displacement vector, so it is also the usual coordinate-distance formula.

Uniform motion over a straight path obeys

$$\text{distance} = \text{speed} \times \text{time}, \qquad \text{average speed} = \frac{\text{total distance}}{\text{total time}}. \tag{F5}$$

For an object at fixed height h observed from a point on level ground, horizontal range is $x = h\cot\theta$.

3. Probability: For equally likely outcomes,

$$P(E) = \frac{n(E)}{n(S)}. \tag{F6}$$

For two events, inclusion–exclusion gives

$$P(A \cup B) = P(A) + P(B) - P(A \cap B), \qquad P(A^c) = 1 - P(A). \tag{F7}$$

When two distinct dice are tossed, the sample space has $6 \cdot 6 = 36$ ordered outcomes; order is retained because the dice are different.

4. Circle geometry and tangents: A radius is perpendicular to the tangent at the point of contact. Tangents drawn from the same external point are equal. The angle between a tangent and a chord equals the angle subtended by that chord in the alternate segment. An angle at the centre is twice an angle at the circumference standing on the same arc.

If a triangle has inradius r and semiperimeter s, then

$$[ABC] = rs, \qquad s = \frac{AB + BC + CA}{2}. \tag{F8}$$

Tangents drawn from the same point to a circle have equal lengths; this lets us express two sides of a circumscribed triangle using the tangent lengths and the in-touchpoint partition of its base.

5. Arithmetic progressions: For first term a and common difference d,

$$t_n = a + (n-1)d, \qquad S_n = \frac{n}{2}\big(2a + (n-1)d\big) = \frac{n}{2}(a + t_n). \tag{F9}$$

Two independent linear relations in a and d can be solved as a matrix system rather than by ad hoc elimination.

6. Coordinate geometry as linear algebra: Represent a point $P(x, y)$ by the column vector $\mathbf{p} = \begin{pmatrix} x \\ y \end{pmatrix}$. Then

$$\overrightarrow{PQ} = \mathbf{q} - \mathbf{p}, \qquad \mathbf{u} \cdot \mathbf{v} = u_x v_x + u_y v_y. \tag{F10}$$

Perpendicular vectors have zero dot product. For $A(x_1, y_1)$, $B(x_2, y_2)$, and $C(x_3, y_3)$, collinearity is equivalent to the vanishing determinant

$$\begin{vmatrix} x_2 - x_1 & y_2 - y_1 \\ x_3 - x_1 & y_3 - y_1 \end{vmatrix} = 0. \tag{F11}$$

The area of a triangle is half the absolute value of the same determinant:

$$[ABC] = \frac{1}{2}\left|\begin{vmatrix} x_2 - x_1 & y_2 - y_1 \\ x_3 - x_1 & y_3 - y_1 \end{vmatrix}\right|. \tag{F12}$$

For a point dividing AB internally in ratio $AP : PB = m : n$,

$$\mathbf{p} = \frac{n\mathbf{a} + m\mathbf{b}}{m + n} = \mathbf{a} + \frac{m}{m+n}(\mathbf{b} - \mathbf{a}). \tag{F13}$$

A 2×2 linear system $M\mathbf{z} = \mathbf{b}$ with nonzero determinant has solution $\mathbf{z} = M^{-1}\mathbf{b}$. This is especially useful for AP data and height–distance systems.

7. Sectors and circular segments: For radius r and central angle θ measured in degrees,

$$A_{\text{sector}} = \frac{\theta}{360^\circ}\pi r^2, \qquad A_{\text{segment}} = A_{\text{sector}} - A_{\triangle}. \tag{F14}$$

For two radii enclosing angle θ, the included triangle has area $\frac{1}{2}r^2 \sin\theta$.

8. Surface area and volume: The formulae needed for this paper are

$$V_{\text{cylinder}} = \pi r^2 h, \qquad A_{\text{curved,cylinder}} = 2\pi r h, \tag{F15}$$

$$V_{\text{cone}} = \frac{1}{3}\pi r^2 h, \qquad A_{\text{curved,cone}} = \pi r \ell, \quad \ell = \sqrt{r^2 + h^2}, \tag{F16}$$

$$V_{\text{sphere}} = \frac{4}{3}\pi r^3, \qquad A_{\text{sphere}} = 4\pi r^2, \tag{F17}$$

$$V_{\text{hemisphere}} = \frac{2}{3}\pi r^3, \qquad A_{\text{curved,hemisphere}} = 2\pi r^2. \tag{F18}$$

In a recasting problem, material volume is conserved. For a cylinder with two hemispherical cavities of radius r, the remaining volume is

$$V_{\text{remaining}} = \pi r^2 h - \frac{4}{3}\pi r^3. \tag{F19}$$

For a cube of side a with a hemisphere of radius r on its top face, the exposed area is $5a^2 + (a^2 - \pi r^2) + 2\pi r^2 = 6a^2 + \pi r^2$.

9. Similarity and construction: If two triangles are similar with corresponding base lengths b_1, b_2, every corresponding linear dimension scales by

$$k = \frac{b_2}{b_1}, \qquad \frac{\text{corresponding side}_2}{\text{corresponding side}_1} = k. \tag{F20}$$

A straight line drawn parallel to one side of a triangle produces a similar triangle (the intercept theorem). This justifies the compass-and-straightedge enlargement used in Q26.

10. A modelling note: Formulae are written here before the questions to make assumptions visible. In the solutions below, a formula is substituted only after its conditions have been checked (for example, a nonzero quadratic coefficient, a right-angle configuration, or an equal-likelihood sample space).

B. Questions and worked solutions

1) Section A – 1 mark each:

Q1. If the quadratic equation $px^2 - 2\sqrt{5}\,px + 15 = 0$ has two equal roots, find p. (CBSE 2015, 30/1, Q1)

Solution: By (F1), the discriminant must be zero:

$$(-2\sqrt{5}p)^2 - 4(p)(15) = 0 \tag{1.1}$$

$$20p^2 - 60p = 0 \tag{1.2}$$

$$20p(p-3) = 0. \tag{1.3}$$

The algebra gives $p = 0$ or $p = 3$. But $p = 0$ removes the x^2 and x terms and leaves $15 = 0$, which is impossible. Therefore

$$\boxed{p = 3}. \tag{1.4}$$

Q2. In Figure 2, tower AB is $20\,\mathrm{m}$ high and shadow BC is $20\sqrt{3}\,\mathrm{m}$ long. Find the Sun's altitude. (CBSE 2015, 30/1, Q2)

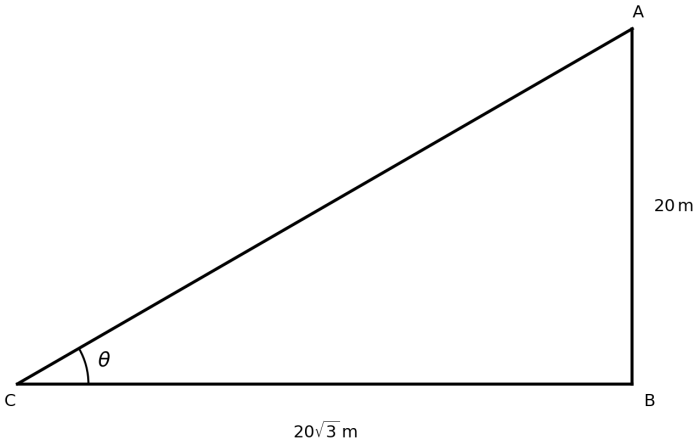


Fig. 2: Official dimensions: height $20\,\mathrm{m}$, shadow $20\sqrt{3}\,\mathrm{m}$.

Solution: Let the angle of elevation be θ. The tower is perpendicular to the ground, so ABC is right-angled at B. Using (F3),

$$\tan\theta = \frac{AB}{BC} = \frac{20}{20\sqrt{3}} = \frac{1}{\sqrt{3}}. \tag{2.1}$$

Hence $\boxed{\theta = 30^\circ}$. The radical in the shadow length is part of the official question; it is not $203\,\mathrm{m}$.

Q3. Two different dice are tossed together. Find the probability that the product of their top numbers is 6. (CBSE 2015, 30/1, Q3)

Solution: By (F6), there are 36 equally likely ordered outcomes. The product is 6 only for

$$(1,6),\ (6,1),\ (2,3),\ (3,2).$$

There are four such outcomes; therefore $P = 4/36 = \boxed{1/9}$.

Q4. In Figure 4, PQ is a chord of a circle with centre O and PT is a tangent. If $\angle QPT = 60^\circ$, find $\angle PRQ$. (CBSE 2015, 30/1, Q4)

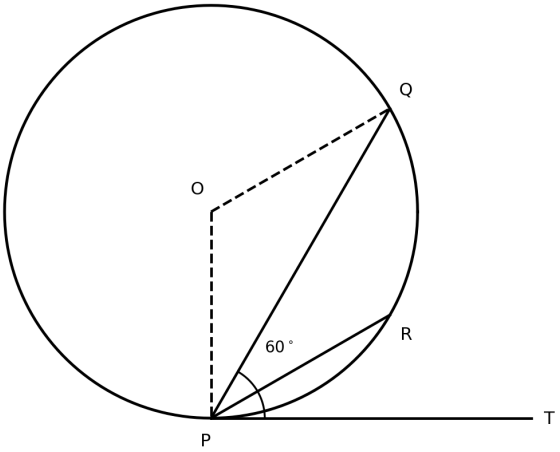


Fig. 4: Chord–tangent configuration as in the official paper.

Solution: Since $OP \perp PT$, $\angle OPQ = 90^\circ - 60^\circ = 30^\circ$. As $OP = OQ$, triangle OPQ is isosceles, so $\angle OQP = 30^\circ$ and $\angle POQ = 120^\circ$. Point R lies on the minor arc PQ, so $\angle PRQ$ subtends the major arc. Its measure is $360^\circ - 120^\circ = 240^\circ$. The angle at the circumference is half its intercepted arc, giving

$$\boxed{\angle PRQ = 120^\circ}.$$

2) Section B – 2 marks each:

Q5. Two tangents RQ and RP are drawn from external point R to a circle with centre O. If $\angle PRQ = 120^\circ$, prove $OR = PR + RQ$. (CBSE 2015, 30/1, Q5)

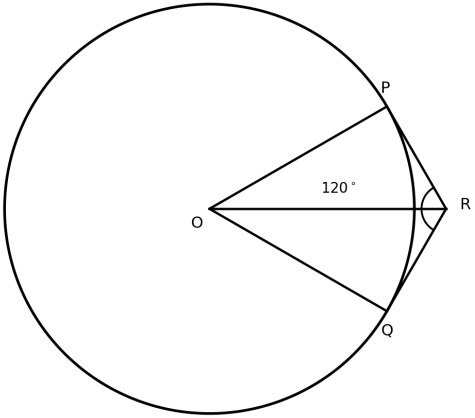


Fig. 5: Two tangents from an external point.

Solution: Radii to tangency points are perpendicular to the tangents, so $\triangle OPR$ and $\triangle OQR$ are right triangles. They have $OP = OQ$ (radii) and common hypotenuse OR, hence are congruent by RHS. Thus $PR = RQ$ and OR bisects $\angle PRQ$, so $\angle PRO = 60°$. In right triangle OPR,

$$\cos 60° = \frac{PR}{OR} = \frac{1}{2} \quad \Longrightarrow \quad OR = 2PR.$$

Since $PR = RQ$, $\boxed{OR = PR + RQ}$.

Q6. In Figure 6, a triangle circumscribes a circle of radius $3\,\mathrm{cm}$. The base portions are $BD = 6\,\mathrm{cm}$ and $DC = 9\,\mathrm{cm}$. If area $[ABC] = 54\,\mathrm{cm}^2$, find AB and AC. (CBSE 2015, 30/1, Q6)

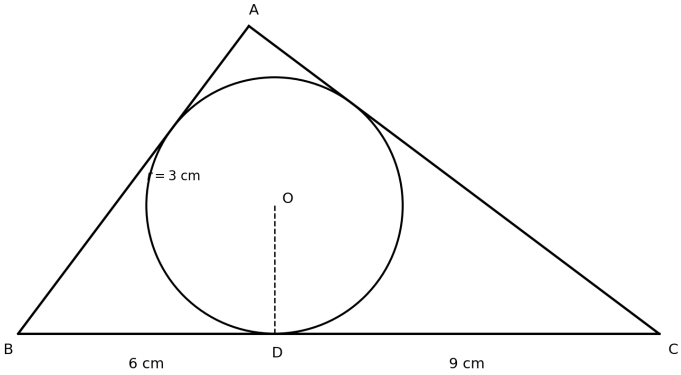


Fig. 6: Circumscribed triangle with the base divided at the tangency point.

Solution: By the inradius relation (F8), the semiperimeter is

$$s = \frac{[ABC]}{r} = \frac{54}{3} = 18\,\text{cm}.$$

Tangents from the same point are equal. If the tangent length from A is x, then $AB = x + 6$, $AC = x + 9$, and $BC = 6 + 9 = 15$. Hence

$$s = \frac{(x+6) + (x+9) + 15}{2} = x + 15 = 18,$$

so $x = 3$. Therefore $\boxed{AB = 9\,\text{cm}, \quad AC = 12\,\text{cm}}$.

Q7. Solve $4x^2 + 4bx - (a^2 - b^2) = 0$ for x. (CBSE 2015, 30/1, Q7)

Solution: Rearranging and using (F2),

$$(2x + b)^2 - a^2 = 0 \quad \Longrightarrow \quad (2x + b - a)(2x + b + a) = 0.$$

Thus $2x + b = \pm a$, giving

$$\boxed{x = \frac{-b + a}{2} \quad \text{or} \quad x = \frac{-b - a}{2}}.$$

Q8. In an AP, $S_5 + S_7 = 167$ and $S_{10} = 235$. Find the AP. (CBSE 2015, 30/1, Q8)

Solution: From (F9),

$$S_5 = 5a + 10d, \quad S_7 = 7a + 21d, \quad S_{10} = 10a + 45d.$$

The conditions become

$$\underbrace{\begin{pmatrix} 12 & 31 \\ 10 & 45 \end{pmatrix}}_{M} \begin{pmatrix} a \\ d \end{pmatrix} = \begin{pmatrix} 167 \\ 235 \end{pmatrix}.$$

Here $\det M = 12(45) - 10(31) = 230 \neq 0$, so the parameters are uniquely determined. Using the inverse matrix,

$$\begin{pmatrix} a \\ d \end{pmatrix} = \frac{1}{230}\begin{pmatrix} 45 & -31 \\ -10 & 12 \end{pmatrix}\begin{pmatrix} 167 \\ 235 \end{pmatrix} = \begin{pmatrix} 1 \\ 5 \end{pmatrix}.$$

The AP is $\boxed{1, 6, 11, 16, \ldots}$.

Q9. $A(4,7)$, $B(p,3)$, and $C(7,3)$ form a right triangle, right-angled at B. Find p. (CBSE 2015, 30/1, Q9)

Solution: The side vectors are $\overrightarrow{BA} = \begin{pmatrix} 4-p \\ 4 \end{pmatrix}$ and $\overrightarrow{BC} = \begin{pmatrix} 7-p \\ 0 \end{pmatrix}$. Perpendicularity means their dot product is zero by (F10):

$$(4-p)(7-p) = 0 \quad \Longrightarrow \quad p = 4 \text{ or } p = 7.$$

The value $p = 7$ makes $B = C$, so no triangle exists. Thus $\boxed{p = 4}$.

Q10. If $A(x,y)$, $B(-5,7)$, and $C(-4,5)$ are collinear, find the relation between x and y. (CBSE 2015, 30/1, Q10)

Solution: By the collinearity determinant (F11),

$$\begin{vmatrix} x+5 & y-7 \\ 1 & -2 \end{vmatrix} = 0.$$

Therefore $-2(x+5) - (y-7) = 0$, or

$$\boxed{2x + y + 3 = 0}.$$

3) Section C – 3 marks each:

Q11. The 14th term of an AP is twice its 8th term. If the 6th term is -8, find the sum of its first 20 terms. (CBSE 2015, 30/1, Q11)

Solution: Let the first term and common difference be a, d. By (F9),

$$a + 13d = 2(a + 7d) \quad \Longrightarrow \quad a + d = 0.$$

Also $a + 5d = -8$. Substituting $a = -d$ gives $4d = -8$, so $d = -2$ and $a = 2$. Thus

$$S_{20} = \frac{20}{2}\big(2(2) + 19(-2)\big) = 10(4 - 38) = \boxed{-340}.$$

Q12. Solve $\sqrt{3}x^2 - 2\sqrt{2}x - 2\sqrt{3} = 0$. (CBSE 2015, 30/1, Q12)

Solution: Here $a = \sqrt{3}$, $b = -2\sqrt{2}$, and $c = -2\sqrt{3}$. Using (F1),

$$\Delta = (-2\sqrt{2})^2 - 4(\sqrt{3})(-2\sqrt{3}) = 8 + 24 = 32.$$

Hence

$$x = \frac{2\sqrt{2} \pm 4\sqrt{2}}{2\sqrt{3}},$$

which gives $\boxed{x = \sqrt{6} \quad \text{or} \quad x = -\frac{\sqrt{6}}{3}}$.

Q13. An aeroplane is observed at elevation 60° from point A. After 15 s, the elevation becomes 30°. At constant height $1500\sqrt{3}\,\text{m}$, find its speed in km/h. (CBSE 2015,

30/1, Q13)

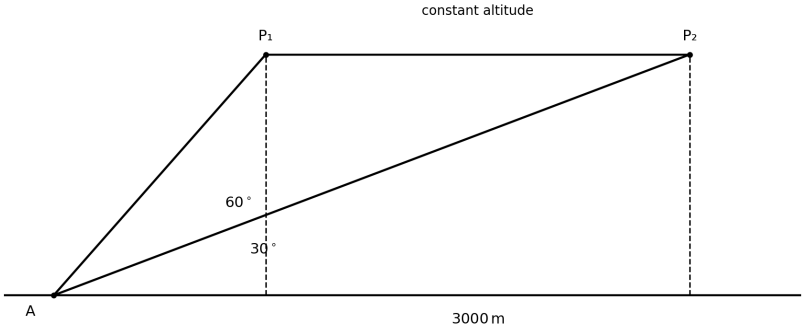


Fig. 13: Horizontal displacement at fixed altitude.

Solution: Let the plane's initial and final horizontal ranges from A be x_1, x_2. Since altitude $h = 1500\sqrt{3}$ is constant, $x = h\cot\theta$:

$$x_1 = 1500\sqrt{3}\cot 60^\circ = 1500\,\text{m}, \qquad x_2 = 1500\sqrt{3}\cot 30^\circ = 4500\,\text{m}.$$

The plane therefore travels $x_2 - x_1 = 3000\,\text{m}$ in 15 s. By (F5),

$$v = \frac{3000}{15} = 200\,\text{m/s} = 200 \times \frac{18}{5} = \boxed{720\,\text{km/h}}.$$

Q14. $A(-2,-2)$ and $B(2,-4)$. Find P on segment AB if $AP = \frac{3}{7}AB$. (CBSE 2015, 30/1, Q14)

Solution: The vector section formula (F13) gives

$$\mathbf{p} = \mathbf{a} + \frac{3}{7}(\mathbf{b}-\mathbf{a}) = \begin{pmatrix}-2\\-2\end{pmatrix} + \frac{3}{7}\begin{pmatrix}4\\-2\end{pmatrix} = \begin{pmatrix}-2/7\\-20/7\end{pmatrix}.$$

Therefore $\boxed{P\left(-\frac{2}{7}, -\frac{20}{7}\right)}$.

Q15. A jar contains only red, blue, and orange balls. $P(R) = \frac{1}{4}$, $P(B) = \frac{1}{3}$, and there are 10 orange balls. Find the total number of balls. (CBSE 2015, 30/1, Q15)

Solution: Let the total be N. The number of red balls is $N/4$ and blue balls is $N/3$. Since these are the only colours, the orange count is

$$N - \frac{N}{4} - \frac{N}{3} = \frac{5N}{12} = 10.$$

Thus $N = 24$, so the jar contains $\boxed{24}$ balls (6 red, 8 blue, and 10 orange).

Q16. Find the minor and major segment areas for a circle of radius $14\,\text{cm}$ and central angle 60°. Use $\pi = 22/7$. (CBSE 2015, 30/1, Q16)

Solution: By (F14), the sector area is

$$A_{\text{sector}} = \frac{60}{360}\pi(14)^2 = \frac{98\pi}{3} = \frac{308}{3}\,\text{cm}^2.$$

The triangle formed by the two radii has included angle 60° and is equilateral with side 14 cm; its area is $\frac{\sqrt{3}}{4}(14)^2 = 49\sqrt{3}\,\text{cm}^2$. Hence

$$A_{\text{minor}} = \boxed{\frac{308}{3} - 49\sqrt{3}\ \text{cm}^2}.$$

The full circle area is $\pi r^2 = 616\,\text{cm}^2$, so

$$A_{\text{major}} = 616 - A_{\text{minor}} = \boxed{\frac{1540}{3} + 49\sqrt{3}\ \text{cm}^2}.$$

Q17. One tent has a cylindrical lower part of diameter $4.2\,\text{m}$ and height $4\,\text{m}$, topped by a cone of the same diameter and height $2.8\,\text{m}$. Canvas costs Rs. 100 per m^2. Find the associations' 50

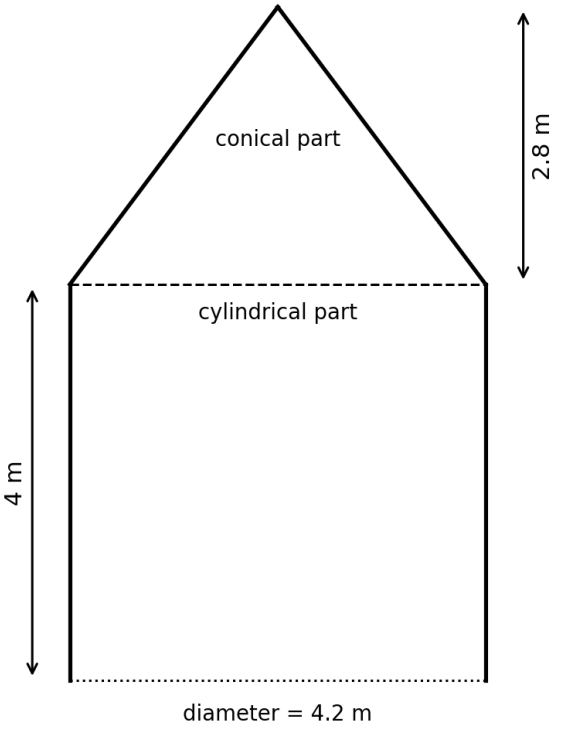


Fig. 17: Tent cross-section: a cylinder plus a cone, open at the bottom.

Solution: The radius is $r = 2.1\,\text{m}$. The cone slant height from (F16) is $\ell = \sqrt{2.1^2 + 2.8^2} = 3.5\,\text{m}$. Canvas area per tent is the sum of the curved areas in (F15) and (F16):

$$A = 2\pi rh + \pi r\ell = 2\pi(2.1)(4) + \pi(2.1)(3.5) = 75.9\,\text{m}^2.$$

For 100 tents, the canvas cost is $100 \times 75.9 \times 100 =$ Rs. 759,000. The promised half is $\boxed{\text{Rs. } 379{,}500}$. The associations show cooperation, initiative, and concern for people affected by floods.

Q18. A hemispherical bowl of internal diameter $36\,\text{cm}$ is filled with liquid. The liquid is poured into 72 cylindrical bottles of diameter $6\,\text{cm}$, with 10

Solution: The bowl radius is 18 cm, so its volume is $\frac{2}{3}\pi 18^3 = 3888\pi\,\text{cm}^3$ by (F18). Only 90

$$72(9\pi h) = 0.9(3888\pi).$$

Cancel π and solve: $648h = 3499.2$, hence $\boxed{h = 5.4\,\text{cm}}$.

Q19. A cube of side $10\,\text{cm}$ is surmounted by a hemisphere. Find the largest possible hemisphere diameter and the cost of painting the exposed solid at Rs. 5 per $100\,\text{cm}^2$; use $\pi = 3.14$. (CBSE 2015, 30/1, Q19)

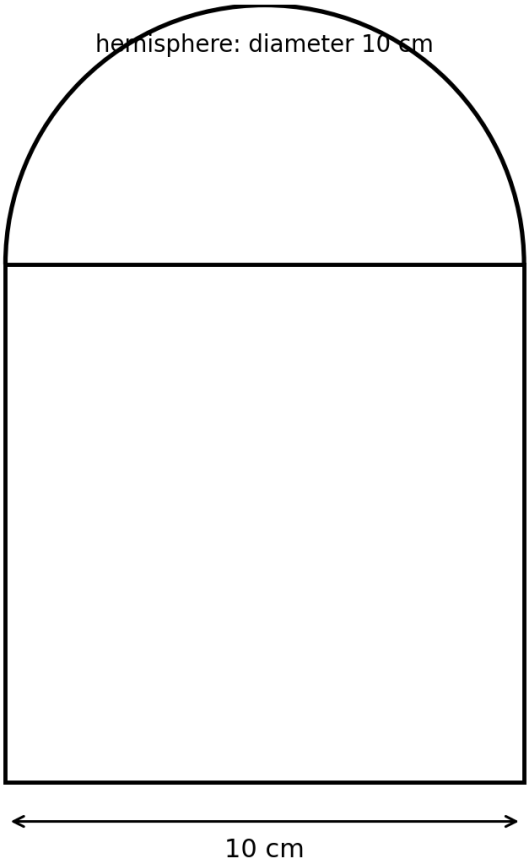


Fig. 19: Hemisphere fits within the cube's top square.

Solution: The largest circular base fitting on a 10 cm square has diameter 10 cm, so $r = 5$ cm. The exposed area equals five complete cube faces, plus the uncovered part of the sixth face, plus the curved hemisphere area:

$$A = 5(10^2) + (10^2 - \pi 5^2) + 2\pi 5^2 = 600 + 25\pi = 678.5\,\mathrm{cm}^2.$$

At Rs. 5 per $100\,\mathrm{cm}^2$, the cost is $678.5 \times 5/100 = \boxed{\text{Rs. } 33.925 \approx \text{Rs. } 33.93}$.

Q20. 504 cones of diameter $3.5\,\mathrm{cm}$ and height $3\,\mathrm{cm}$ are melted and recast into a sphere. Find its diameter and surface area; use $\pi = 22/7$. (CBSE 2015, 30/1, Q20)

Solution: Each cone has radius 1.75 cm, and the total cone volume is

$$504\left(\frac{1}{3}\pi(1.75)^2(3)\right) = 1543.5\pi\,\mathrm{cm}^3.$$

Equating this to the sphere volume from (F17),

$$\frac{4}{3}\pi R^3 = 1543.5\pi \quad \Longrightarrow \quad R^3 = 1157.625 = (10.5)^3.$$

Thus the diameter is $\boxed{21\,\text{cm}}$. Its surface area is $4\pi(10.5)^2 = 441\pi = \boxed{1386\,\text{cm}^2}$.

4) Section D – 4 marks each:

Q21. A rectangular field's diagonal is 16 m longer than its shorter side. Its longer side is 14 m longer than its shorter side. Find the side lengths. (CBSE 2015, 30/1, Q21)

Solution: Let the shorter side be $x > 0$. The longer side is $x + 14$ and diagonal $x + 16$. Pythagoras gives

$$x^2 + (x+14)^2 = (x+16)^2 \quad \Longrightarrow \quad x^2 - 4x - 60 = 0.$$

Factor: $(x-10)(x+6) = 0$. The positive length is $x = 10$ m, and the longer side is 24 m. Therefore $\boxed{10\,\text{m by } 24\,\text{m}}$ (diagonal 26 m).

Q22. Find the 60th term of the AP $8, 10, 12, \ldots$ and the sum of its last 10 terms. (CBSE 2015, 30/1, Q22)

Solution: Here $a = 8$ and $d = 2$. By (F9),

$$t_{60} = 8 + 59(2) = \boxed{126}.$$

The 51st term is $t_{51} = 8 + 50(2) = 108$. Therefore the last ten terms form an AP with first term 108 and last term 126; their sum is

$$S_{\text{last }10} = \frac{10}{2}(108 + 126) = \boxed{1170}.$$

Q23. A train travels 54 km at speed x km/h and then 63 km at $(x+6)$ km/h. Total journey time is 3 h. Find x. (CBSE 2015, 30/1, Q23)

Solution: Using time $=$ distance/speed from (F5),

$$\frac{54}{x} + \frac{63}{x+6} = 3.$$

Multiplying through by $x(x+6)$ gives $54(x+6) + 63x = 3x(x+6)$, hence

$$x^2 - 33x - 108 = 0 = (x-36)(x+3).$$

Only a positive speed is admissible, so $\boxed{x = 36\,\text{km/h}}$.

Q24. Prove that tangent lengths drawn from an external point to a circle are equal. (CBSE 2015, 30/1, Q24)

Solution: Let PA and PB touch the circle at A and B, with centre O. Join OA, OB, and OP. Since a radius is perpendicular to a tangent, $\angle OAP = \angle OBP = 90^\circ$. Also $OA = OB$ (radii) and OP is common. Hence $\triangle OAP \cong \triangle OBP$ by RHS, so corresponding sides satisfy

$$\boxed{PA = PB}.$$

Q25. Prove that the tangent at the midpoint of an arc is parallel to the chord joining the arc's endpoints. (CBSE 2015, 30/1, Q25)

Solution: Let M be the midpoint of arc AB and O be the centre. Because arcs

AM and MB are equal, their chords are equal: $MA = MB$. In isosceles triangle AMB, the line OM through the arc midpoint is the perpendicular bisector of AB, so $OM \perp AB$. The tangent at M is perpendicular to radius OM as well. Two lines perpendicular to the same line are parallel; therefore

$$\boxed{\text{tangent at } M \parallel AB}.$$

Q26. Construct $\triangle ABC$ with $AB = 6$ cm, $\angle A = 30^\circ$, $\angle B = 60^\circ$. Then construct a similar triangle $AB'C'$ with $AB' = 8$ cm. (CBSE 2015, 30/1, Q26)

Solution:

(a) Draw $AB = 6$ cm. At A, construct a 30° ray to AB; at B, construct a 60° ray to BA. Their intersection is C.

(b) From A, draw an auxiliary ray AX not collinear with AB. Mark eight equal intervals on it to obtain $A_1, \ldots, A_8$.

(c) Join A_6 to B. Through A_8, draw a line parallel to A_6B; let it meet the ray AB produced at B'. By the intercept theorem, $AB'/AB = AA_8/AA_6 = 8/6$, hence $AB' = 8$ cm.

(d) Through B', draw a line parallel to BC to meet the ray AC produced at C'. Then $\triangle AB'C' \sim \triangle ABC$ by corresponding angles, and the required construction is complete.

Q27. Point A is 20 m above a lake. The elevation angle of a cloud is 30°, and the depression angle of its reflection is 60°. Find the distance from A to the cloud. (CBSE 2015, 30/1, Q27)

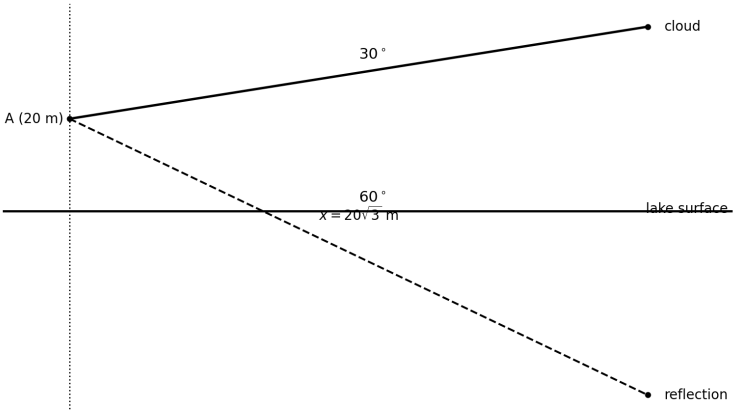


Fig. 27: Cloud and its reflection are symmetric about the lake surface.

Solution: Let the cloud height above water be H and the horizontal distance be x. The two angle conditions give

$$\frac{H-20}{x} = \tan 30^\circ = \frac{1}{\sqrt{3}}, \qquad \frac{H+20}{x} = \tan 60^\circ = \sqrt{3}.$$

Equivalently, a linear system in (H, x) is

$$\begin{pmatrix} 1 & -1/\sqrt{3} \\ 1 & -\sqrt{3} \end{pmatrix} \begin{pmatrix} H \\ x \end{pmatrix} = \begin{pmatrix} 20 \\ -20 \end{pmatrix}.$$

Solving gives $H = 40$ m and $x = 20\sqrt{3}$ m. Since the cloud is $H - 20 = 20$ m above A vertically,

$$AC = \sqrt{x^2 + (H-20)^2} = \sqrt{(20\sqrt{3})^2 + 20^2} = \boxed{40\,\text{m}}.$$

Q28. A card is drawn from a well-shuffled 52-card deck. Find the probability that it is (i) a spade or ace, (ii) a black king, (iii) neither a jack nor a king, (iv) either a king or a queen. (CBSE 2015, 30/1, Q28)

Solution: By (F6) and (F7):

(i) 13 spades and 4 aces overlap at the ace of spades, so $P = (13+4-1)/52 = \boxed{4/13}$.

(ii) There are two black kings, so $P = 2/52 = \boxed{1/26}$.

(iii) Four jacks and four kings are disjoint, so $P = (52-4-4)/52 = \boxed{11/13}$.

(iv) There are four kings and four queens, so $P = (4+4)/52 = \boxed{2/13}$.

Q29. Find k if the triangle with vertices $(1, -1)$, $(-4, 2k)$, and $(-k, -5)$ has area 24 square units. (CBSE 2015, 30/1, Q29)

Solution: By the determinant area formula (F12),

$$2[ABC] = \left\| \begin{matrix} -5 & 2k+1 \\ -k-1 & -4 \end{matrix} \right\| = \left| 2k^2 + 3k + 21 \right|.$$

Since the quadratic has positive leading coefficient and negative discriminant, it is positive for every real k. Thus $2k^2 + 3k + 21 = 48$, or

$$2k^2 + 3k - 27 = 0 = (2k+9)(k-3)$$

Solving directly gives discriminant 225, hence $k = (-3 \pm 15)/4$. Therefore

$$\boxed{k = 3 \quad \text{or} \quad k = -\frac{9}{2}}.$$

Q30. A square lawn has side 42 m. Two circular flower beds lie outside the vertical sides, each circle centred at the square's diagonal intersection O. Find the total shaded area. (CBSE 2015, 30/1, Q30)

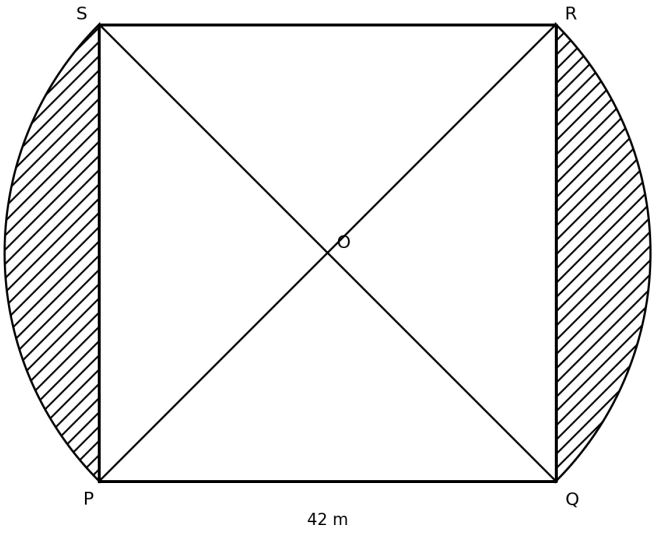


Fig. 30: Two congruent circular segments outside the square.

Solution: The centre-to-corner radius is half the diagonal, so

$$r^2 = \frac{42^2 + 42^2}{4} = 882\,\mathrm{m}^2.$$

Each side subtends 90° at the centre. By (F14), one segment's area is $(90/360)\pi r^2 - \frac{1}{2}r^2 = \frac{1}{4}\pi(882) - 441$. There are two equal beds, hence

$$A_{\text{total}} = 2\left(\frac{882\pi}{4} - 441\right) = 441\pi - 882.$$

With $\pi = 22/7$, $A_{\text{total}} = 1386 - 882 = \boxed{504\,\mathrm{m}^2}$.

Q31. From both ends of a metal cylinder, hemispherical cavities of the same diameter are scooped out. The cylinder has height 10 cm and radius 4.2 cm. The remaining metal is made into a cylindrical wire of thickness 1.4 cm. Find the wire length; use $\pi = 22/7$. (CBSE 2015, 30/1, Q31)

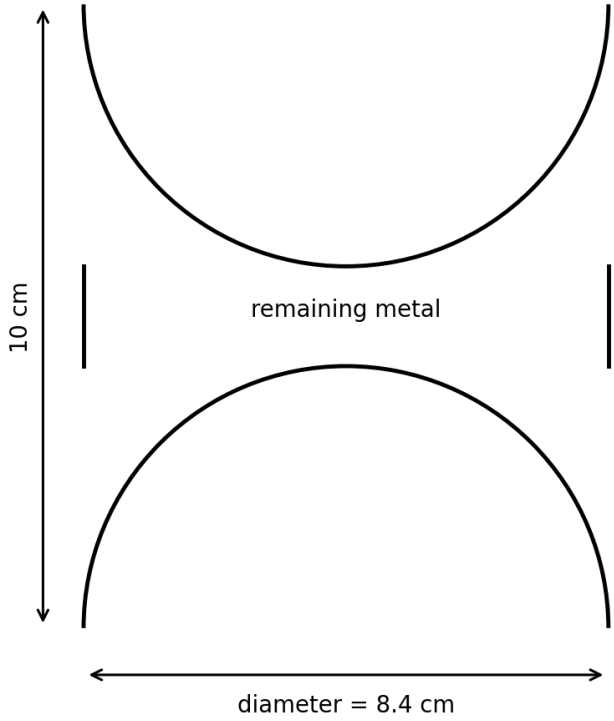


Fig. 31: The removed pair of hemispheres has the volume of one sphere.

Solution: The two hemispherical cavities together remove the volume of one sphere. From (F19), the remaining volume is

$$V = \pi(4.2)^2(10) - \frac{4}{3}\pi(4.2)^3 = 77.616\pi\,\mathrm{cm}^3.$$

A wire thickness of 1.4 cm means radius 0.7 cm. Equating the remaining volume to the wire volume $\pi(0.7)^2 L$ gives

$$L = \frac{77.616\pi}{0.49\pi} = \boxed{158.4\,\mathrm{cm}}.$$

Numerical and transcription notes: The determinant in Q29 is $2k^2+3k+21$; its correct factorization after setting the area to 24 is $2k^2 + 3k - 27 = (2k + 9)(k - 3)$. Therefore the two values are 3 and $-9/2$. The build-time Python checks independently recompute key outputs and ensure every generated figure exists.